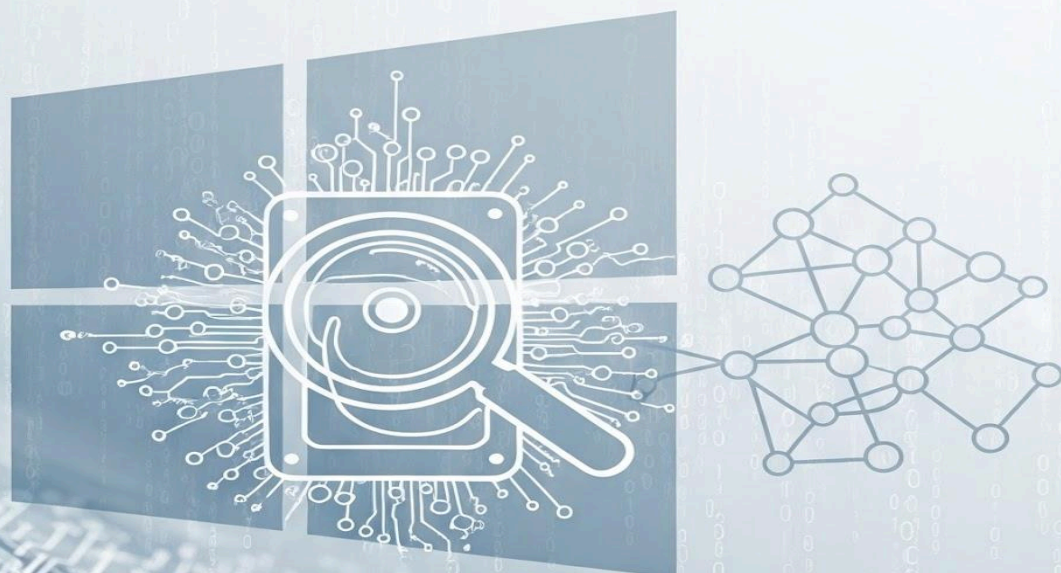


Windows Forensics

Program Code: NX212

Student: S12 (TMagen773640)

Yonatan Menashe



Analyzer

Automate HDD and Memory Analysis

Vision & Mission

Project Goal

The goal of this project is to automate the forensic analysis of Windows evidence files. Instead of running several forensic tools manually, this script performs the entire investigation workflow automatically.

The tool can analyze both disk images and memory dumps, extract artifacts, run memory analysis with Volatility, and generate a final report summarizing the investigation results.

The Mission

The mission of this project is to simplify the forensic investigation process by combining multiple forensic tools into a single automated workflow. Instead of manually running different tools and collecting results separately, the script performs artifact extraction, memory analysis, and reporting automatically. This approach makes the investigation faster, more consistent, and easier to reproduce during forensic analysis.

How it Works

The user provides a forensic evidence file such as a disk image or memory dump. The script automatically runs several forensic tools, extracts artifacts, performs memory analysis when possible, and generates a final investigation report.

Technical Walkthrough

To keep the investigation process organized, the script is divided into several functions where each function performs a specific stage of the forensic workflow.

check_root

This function verifies that the script is running with root privileges, which are required for installing tools and performing forensic analysis.

check_file

The script asks the user to provide an evidence file and validates that the file exists before starting the analysis.

install_tools

This function checks whether the required forensic tools are installed on the system and installs them automatically if they are missing.

run_carvers

During this stage, the script runs several forensic utilities such as Foremost, Bulk Extractor, Binwalk, Strings, and Exiftool to extract artifacts from the evidence file.

memory_analysis

If the provided evidence is a memory dump, the script uses Volatility plugins to analyze processes, registry keys, command history, and network connections.

generate_report

After the analysis is complete, the script generates a summary report that includes information about the analyzed file and the extracted artifacts.

zip_results

Finally, the script compresses the analysis results into a ZIP archive for easier storage and review.

Execution Evidence

1. The script starts by verifying that it is running with root privileges and then prompts the user to provide an evidence file for analysis. This ensures that the investigation begins with a valid forensic evidence file.

```
(root@kali)-[/home/kali/project]
# ./forensics_project.sh
== Automated Windows Forensics Tool ==
The script is running with root privilege
Enter file to analyze: 
```

2. After the user provides the evidence file, the script identifies the file type and verifies that the file exists. This step ensures that the analysis begins with valid forensic evidence before continuing to the next stage.

```
Enter file to analyze: memdump.mem
File found: memdump.mem
File Type: data
```

3. After validating the evidence file, the script verifies that the required forensic tools are installed on the system. If any tool is missing, the script installs it automatically so the analysis can continue.

```
Checking for tools..
binwalk is installed
bulk_extractor is installed
foremost is installed
strings is installed
exiftool is installed
voli is installed
```

4. The script creates a dedicated output directory for the investigation. All extracted files, analysis results, and reports are stored in this directory to keep the evidence organized.

```
Output directory created: /home/kali/project/analysis_2026-03-10_11-40-17
Carving and strings analysis completed
```

```
(root@kali)-[/home/kali/project/analysis_2026-03-10_11-40-17]
# ls -l
total 8124
drwxr-xr-x 2 root root 4096 Mar 10 11:41 binwalk
-rw-r--r-- 1 root root 1172 Mar 10 11:41 binwalk.txt
drwxr-xr-x 7 root root 4096 Mar 10 11:41 bulk
-rw-r--r-- 1 root root 440 Mar 10 11:41 exif_metadata.txt
drwxr-xr-x 6 root root 4096 Mar 10 11:41 foremost
-rw-r--r-- 1 root root 160395 Mar 10 11:41 human_hits.txt
-rw-r--r-- 1 root root 310 Mar 10 11:41 report_2026-03-10_11-40-17.txt
-rw-r--r-- 1 root root 8125979 Mar 10 11:41 strings.txt
drwxr-xr-x 2 root root 4096 Mar 10 11:41 voli
```

5. Then the script searches the extracted data for potential network traffic artifacts. If PCAP files are found, the script displays their location and size so they can be analyzed later as part of the investigation.

```
Searching for network traffic artifacts ...
Network traffic found:
PCAP: /home/kali/project/analysis_2026-03-10_11-40-17/bulk/packets.pcap (Size: 104K)
```

6. Next, the script performs memory analysis using Volatility. The tool automatically detects the correct memory profile and runs several plugins to analyze system processes, command history, registry keys, and network connections.

```
Starting memory analysis ...
Detected memory profile: WinXPSP2x86
Running plugin: pstree
Running plugin: printkey
Running plugin: pslist
Running plugin: cmdline
Running plugin: connscan
```

7. Finally, the script generates a forensic report summarizing the analysis results and then compresses the investigation directory into a ZIP archive for easier storage and review.

```
Generating report ...
Report created: /home/kali/project/analysis_2026-03-10_11-40-17/report_2026-03-10_11-40-17.txt
Creating archive ...
Archive created: analysis_2026-03-10_11-40-17.zip
```

Generated Forensic Report:

```
(root@kali)-[/home/kali/project/analysis_2026-03-10_11-40-17]
# cat report_2026-03-10_11-40-17.txt
===== FORENSICS ANALYSIS REPORT =====
Analysis Time: 2026-03-10_11-40-17
Analysis Duration: 90 seconds
Analyzed File: memdump.mem
File Size: 513M
File Type: data
Output Directory: /home/kali/project/analysis_2026-03-10_11-40-17
Total Extracted Files: 16012
Volatility Output Files: 5
Human Readable Hits: 5023
```

Conclusion

The project shows how a forensic investigation workflow can be automated using a Bash script.

The tool combines multiple forensic utilities such as Foremost, Bulk_extractor, Binwalk, Strings, Exiftool, and Volatility into a single automated process.

By automating artifact extraction, memory analysis, and reporting, the tool makes the investigation process faster and easier.

The generated report and archived results allow investigators to quickly review the findings and preserve the analysis results for further examination.